

STATISTICS

PART 2: Measures of Central Tendency (SUPPLEMENTARY DOCUMENT)

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Measures of Central Tendency

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Properties of Geometric Mean

1. **Definition:**

For n positive numbers $x_1, x_2, \dots, x_n$

$$\overline{x_g} = \left(\prod_{i=1}^n x_i \right)^{\frac{1}{n}}$$

2. **Applicability:**

Defined only for positive values

3. **Less sensitive to outliers:**

The geometric mean reduces the effect of extreme (especially large) values compared to the arithmetic mean.

4. **Not additive (not linear):**

Unlike the arithmetic mean, the geometric mean is not additive

$$GM(x + y) \neq GM(x) + GM(y)$$

5. **Multiplicative nature:**

It is appropriate for data that combine **multiplicatively** (e.g., average growth rates, average interest rates, average population increases).

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Example: An investment grows by 50% in the first year and then decreases by 50% in the second year. What is the **mean of annual growth rate** over the two years?

Solution:

The growth factors are 1.5 and 0.5 for grows by 50% and then decreases by 50%, respectively. Let the initial value be \$100.

- After the first year: $100 \times 1.5 = 150$
- After the second year: $150 \times 0.5 = 75$

So after two years, the investment is worth **\$75 — a 25% total loss.**

Geometric Mean (correct)

mean of annual
growth rate

$$\overline{x}_g = \left(\prod_{i=1}^n x_i \right)^{\frac{1}{n}} = (1.5 \times 0.5)^{\frac{1}{2}} \cong 0.866 \quad \longrightarrow \quad 0.866 - 1 = -0.134 = -13.4\%$$

$100 \times 0.866 \times 0.866 \cong 75$ **\$75 — a 25% total loss**

Arithmetic Mean (incorrect)

$$\overline{x}_a = \frac{1}{n} \sum_{i=1}^n x_i = \frac{1.5 + 0.5}{2} = 1 \quad (\text{incorrect})$$

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Properties of Harmonic Mean

1. Definition:

For n positive numbers $x_1, x_2, \dots, x_n$

$$\overline{x_h} = \frac{n}{\sum_{i=1}^n \frac{1}{x_i}}$$

2. Applicability:

Defined only for positive values

3. Insensitive to large outliers:

Large outliers can dramatically affect the result.

4. Not additive (not linear):

Unlike the arithmetic mean, the harmonic mean is not additive

$$HM(x + y) \neq HM(x) + HM(y)$$

5. Multiplicative nature:

Emphasizes smaller values more than larger ones, because it takes the reciprocal of each data point.

6. Use case:

Used when averaging **rates**, **ratios**, or **speeds** — especially when the quantity involved is constant (equal distance, equal work, equal amount). **Ex:** Average speed over equal distances, average production rate for equal quantities

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Example: A car travels the same distance twice, first at 60 km/h and then at 120 km/h. What is the **mean of the speed** of over the whole trip?

Solution:

Let each distance be 60 km.

- Segment 1: 60 km at 60 km/h
- Segment 2: 60 km at 120 km/h

The total distance is 120 km and the total travel time is 1.5 (1+0.5) hours. So, the mean of the speed is $120/1.5=80$ km/h

Harmonic Mean (correct)

$$\overline{x}_h = \frac{n}{\sum_{i=1}^n \frac{1}{x_i}} = \frac{2}{\frac{1}{60} + \frac{1}{120}} = \frac{2}{\frac{2}{120} + \frac{1}{120}} = 80 \text{ km/h}$$

Arithmetic Mean (incorrect)

$$\overline{x}_a = \frac{1}{n} \sum_{i=1}^n x_i = \frac{60 + 120}{2} = 90 \text{ km/h} \quad (\text{incorrect})$$

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Example: A car travels for **equal amounts of time** at two different speeds:

- For half of the trip time, it drives at **60 km/h**.
- For the other half, it drives at **120 km/h**.

What is the **mean speed** of the car over the whole trip?

Arithmetic Mean (correct)

$$\bar{x}_a = \frac{1}{n} \sum_{i=1}^n x_i = \frac{60 + 120}{2} = 90 \text{ km/h}$$

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Properties of Quadratic Mean (Root Mean Square-RMS)

1. Definition:

For n real numbers $x_1, x_2, \dots, x_n$

$$\overline{x_q} = \sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2}$$

2. Applicability:

Defined both positive and negative values.

3. Sensitive to large outliers:

Large outliers dramatically affect the result.

4. Not additive (not linear):

Unlike the arithmetic mean, the quadratic mean is not additive

$$\text{RMS}(x + y) \neq \text{RMS}(x) + \text{RMS}(y)$$

5. Relationship with variance:

$$\overline{x_q}^2 = \overline{x_a}^2 + \sigma_X^2$$

6. Use case:

Commonly used when the **magnitude** (not sign) of a quantity matters **Ex:** voltage, current, waves, or error magnitude (Root-mean-square error (RMSE) in regression or machine learning)

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Example: The following data represent the number of defective items found in 7 daily samples of a production line:

2, 3, 4, 6, 8, 10, 12

Find the **mean deviation from the median** of this dataset.

Solution:

Me=6

deviations from the median : $\{2-6, 3-6, 4-6, 6-6, 8-6, 10-6, 12-6\}$
 $= \{-4, -3, -2, 0, 2, 4, 6\}$

Quadratic Mean (correct)

$$\overline{x_q} = \sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2} = \sqrt{\frac{1}{7} ((-4)^2 + (-3)^2 + (-2)^2 + (0)^2 + (2)^2 + (4)^2 + (6)^2)} \cong 3.48$$

Arithmetic Mean (incorrect)

$$\overline{x_a} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{(-4) + (-3) + (-2) + 0 + 2 + 4 + 6}{7} \cong 0.43$$